MILESTONE 2 · MODEL AND METHOD INTEGRATION

AI Inventory Replenishment Planner

Forecasting, EOQ and reorder points on real H&M sales, explained by a conversational AI agent



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DATA

One year of real H&M daily sales

31M
purchases of 105,000 products (Kaggle)

40
steadiest sellers, sold on ≥ 90% of days

18
products after cleaning, 365 days each

Cleaning rules

- No gaps, negatives or duplicate days
- Dropped 11 products being phased out (last 4 weeks < 50% of their yearly average)
- At most 3 per product type, for variety

Not in the data

- Stock levels: simulated (3 to 12 days of real sales), since real stock is not public
- Costs and lead time: labelled assumptions the user can change

Why 18 products?

- Only steady, active sellers make forecasting meaningful
- The same code scales to any number of products

Source: H&M Personalized Fashion Recommendations (Kaggle), 2019-09-24 to 2020-09-22

INPUTS

What the models need, and how each is checked

Input	Source	Validation
Daily sales history	H&M data	365 days, no gaps or negatives
Current stock	Simulated (seed 42)	Non-negative number per product
Ordering cost S	Assumption: $50	Number, greater than 0
Holding cost H	Assumption: $2/unit/year	Number, greater than 0
Lead time L	Assumption: 7 days	Greater than 0, at most 365
Demand change	User scenario	At least -100%
Forecast window	User setting: 7 days	One of 3, 7, 14 or 28
Product	User's words	Name, type or ID; ambiguous → asks

Safety stock (optional): service level must be 0, 90, 95, 97.5 or 99%

COLLECTING AND CHECKING INPUTS

The agent asks instead of guessing

“What happens if demand increases?”	**Missing input:** asks by what percentage
“Set lead time to -5 days”	**Invalid input:** lead time must be greater than 0; asks again
“How is the t-shirt doing?”	**Ambiguous:** matches Brit Baby Tee, Cat Tee and Therese tee; asks which

Validation lives in Python, so the dashboard form and the chat follow the same rules. Errors go back to the agent, which explains them in plain words.

User message (dark) · agent response (light)

HOW THE AGENT CALLS THE MODELS

The AI chooses the tool; Python does the math

1. Question
"What if demand rises 20%?"

→

2. Gemini LLM
Picks run_what_if, demand +20

→

3. Python models
Validate, recalculate, return JSON

→

4. Explanation
Units 5,048 → 5,962, and why

Eight tools the agent can call

get_inventory_summary · list_products · get_product_details · prioritize_reorders · run_what_if · update_assumptions · reset_assumptions · get_assumptions

Every answer shows a **Tools used** panel with each call and its result, and a **numbers checked** line confirming every number came from Python.

Built with Python, Streamlit and the Google Gemini API (function calling)

PRELIMINARY RESULTS

8 of 18 products need reordering now

18

products analysed

8

need reorder (stock ≤ ROP)

5,048

units recommended in total

Most urgent first: fewest days of stock left



1. Harrison top

3.6 days left · order 704



2. Milk RW slack

3.8 days left · order 474



3. Lucy blouse

4.1 days left · order 891

Defaults: S = $50, H = $2/unit/year, lead time 7 days, 7-day forecast, no safety stock

TESTING

29 automated tests, all passing

2	9	8	10
Data	**Formulas**	**Validation**	**Agent answers**
365 days, no gaps, a photo per product, baseline totals	Hand-checked SMA, MAE, EOQ, ROP, safety stock, scenarios	Negative, zero, text and out-of-range inputs; unknown or ambiguous products	Asks for missing values, rejects bad inputs, catches invented numbers

Hand check: Brit Baby Tee sold 65 units in 7 days → ROP 65, EOQ 412, stock 65 ≤ 65 → reorder.
Live AI test: 6 of 6 demo answers from Gemini matched the Python model. Every answer in the app now shows an automatic check that its numbers came from Python.

python -m pytest · tests/test_inventory.py

LIVE DEMO

From question to recommendation

Step	We ask	The agent
1. Objective	Which products should I prioritize?	Ranks Harrison top first
2. Missing input	What happens if demand increases?	Asks: by what percentage?
3. Check input	Set lead time to -5 days	Explains it must be above 0
4. Run model	What if demand increases by 20%?	Units 5,048 → 5,962
5. Explain	Why does Brit Baby Tee need replenishment?	Stock 65 ≤ ROP 65, order 412
6. Revise	Change lead time to 14 days	Dashboard: all 18 reorder

ai-inventory-replenishment-planner-4dtk2bp4nqeuo73l8pqlan.streamlit.app

REMAINING CHALLENGES

What we are working on next

Challenges

Validate the live agent
Run the demo questions with real Gemini responses

Simulated stock and costs
H&M publishes sales only, not inventory or costs

Moving averages lag trends
Lucy blouse sales more than doubled in the final weeks

Plan for the final

Second forecast model
Exponential smoothing, best model chosen per product

Upload your own stock
Let users enter real stock levels and costs

Evaluation write-up
Test cases, report and team contributions

[Edit to match your team's actual plan]

Thank you

Questions?

Live app: ai-inventory-replenishment-planner-4dtk2bp4nqeuo73l8pqlan.streamlit.app

Code: github.com/QuantumFluxx-pew/AI-Inventory-Replenishment-Planner

Uses public H&M data · Not affiliated with H&M